Adobe After Effects Basics

Adobe After Effects is an amazing motion graphics program that can be used to create interesting animation and video graphics. In this lab we will learn the basics of setting up an Adobe Effect project, creating a composition, importing files and applying basic motion property animations.

Setup

Take a moment to look at the video will will create using Adobe After Effects. You can view the video on YouTube below:

https://youtu.be/Sf9Op-6ircE

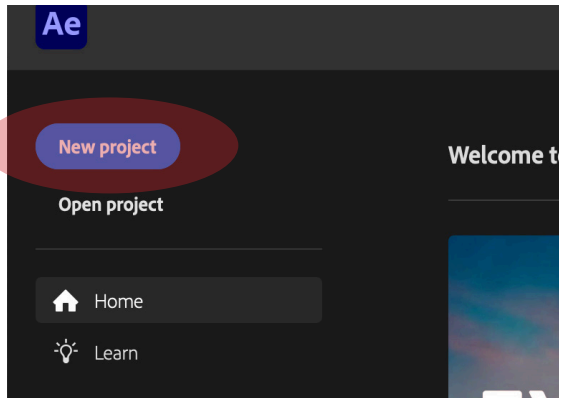
Make sure you have downloaded the lab files needed to complete this lab. The files are in zip folder called *After-Effects-Basics.zip*. Make sure to unzip the folder completely to use the files inside appropiately.

After Effects Overview

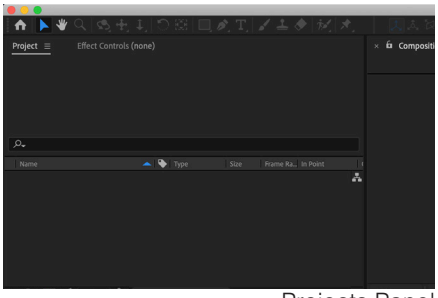
Begin by loading the application Adobe After Effects on your computer.

Before we can begin we'll need to create a project file. The project file will store all of the references files like video, images and audio, as well as store our timeline animation.

Create a new project by clicking the *New Project* option on the home screen for the program. This will take you to main Adobe After Effects layout that looks like the image on the right.

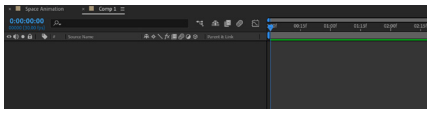


Let's begin by going over the basic user interface of After Effects. At the top left is the *Project* tab. The project tab lists all of the content we can use as part of our project. This includes imported video, imported audio, imported images and compositions as well.



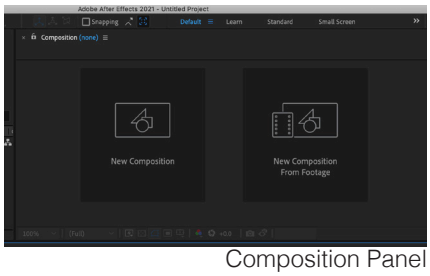
Projects Panel

Below the *Project* tab is the *Timeline*. The *timeline* represents our composition throughout time, meaning beginning, middle and end. We will eventually place content like images, audio and video into the timeline. The timeline is also where we will eventually animate our content.



Timeline Panel

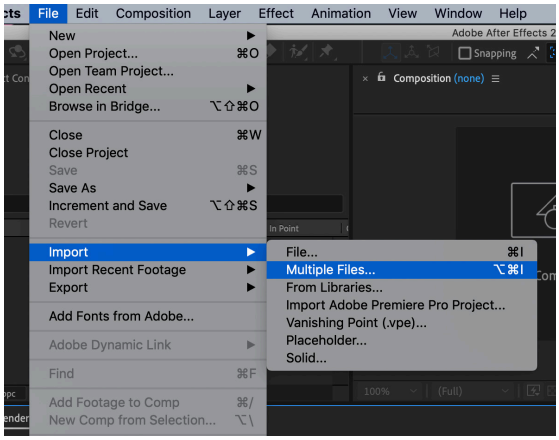
To the right the *Project* section, above the *Timeline* section is the *Composition* section. The composition section allows us to view our timeline. This allows us to see the "final" project before we export it to a movie format. The *Composition* section is also where we will edit the content for time and length.



Composition Panel

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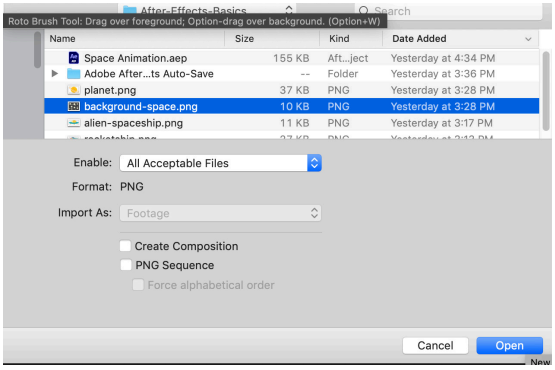
Import Files into Adobe After Effects



In this project we will be creating an animated space scene. In order to do this will need to import content / files into our project. Most content we work will in Adobe After Effects will not be made by the program itself, but instead will be imported from our project folder.

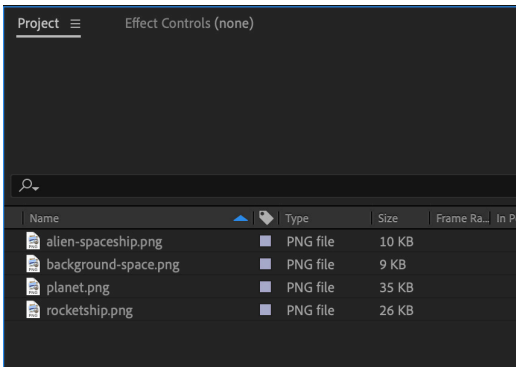
To import files go to *File > Import > Multiple Files*. Selecting this will open a dialogue box like the one on the right. Select the following file and choose *Open.*

Space-Background.png



This is an image file we will use in our project. Since we choose *Multiple Files* we can import multiples files at a time, instead of just one. Select the following files and use the *Open* button.

- *Rocketship.png*
- *Planet.png*
- *Alien-Spaceship.png*

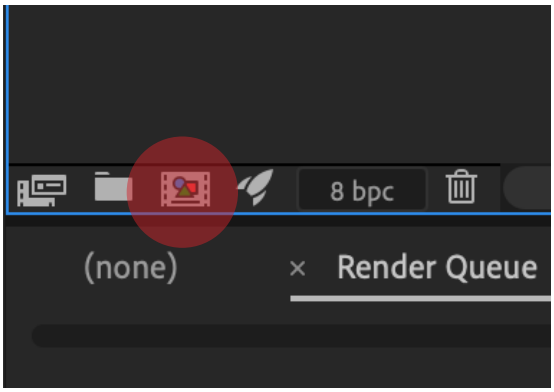


To finish importing files hit the *Cancel* button. After importing the files you should see the files appear in the *Project* tab. The files will be listed alphabetically and you can view other information about each file including file format, file size and more.

Now that we've done a bit of work, let's save our After Effects project file. Go to *File > Save* to save your project. The native format of After Effects is the After Effects Project file or .aep file. Save the project as *lastname-after-effects-basics.aep*, putting in your lastname for *lastname*. Save the project to the lab project folder containing the images.

Create a Composition

Now that we have files to work with, we need a composition to animate them in. A composition is the file that stores our files and the animation we apply to it.

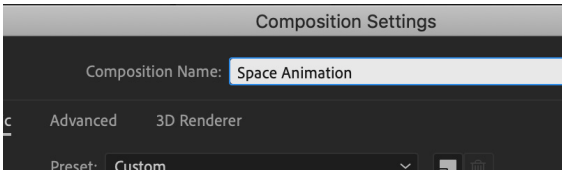


Let's create a composition for our space animation project. At the bottom of the Projects tab click the *"Create a New Composition"* button, the button looks like a piece of film footage. Clicking the button this will create a *Composition Settings* dialogue box to appear. This dialogue box is where we will set the settings of our composition.

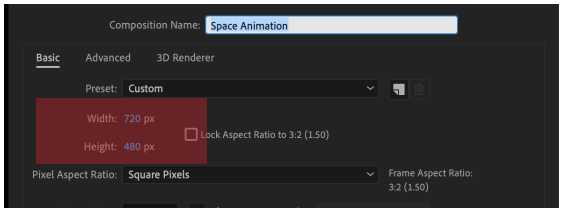
When determining the settings of your composition it is important to think of the composition file like the canvas for a painter. Whatever the settings of your composition file are what the final animation video will look like.

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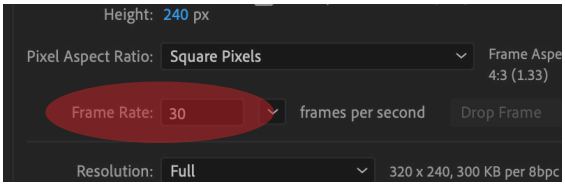
Begin by giving a name to the composition. This is the name that will eventually appear in the *Projects* tab, let's call our composition *Space Animation*.



The first major setting options is the size of our composition. The dimensions we set will be the dimensions of our final video. Typically there would be certain standard sizes we may want to set it to such as standard HD or film size. For this lab we will use a smaller dimension size to make the file easier to work with. Set the *width* and *height* to 720px x 480px.



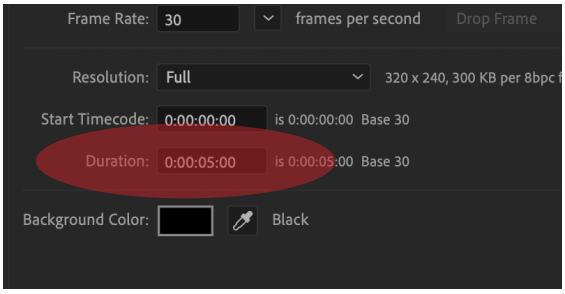
The next important aspect of the *Composition Settings* is the *Frame Rate* option. The frame rate is how many frames of the animation will be played per second. For television and film there are standard frate rates used by the industry. For this lab we will use 30 frames per second.



The next important element of the *Composition Settings* is the aspect ratio option. This should be kept at Square pixels for now.

Finally, we are going to set the *duration* of the composition. The *duration* is how long the timeline for are animation will be. Let's set this to five seconds. After Effects uses what is referred to as *timecode*, timecode is how we represent where we are in time while editing. Your time code would look like this

00:00:05:00



The green number represent *hours*, the blue number represents *minutes*, the red number represent *seconds* and the purple number represents *fractions of a second*.

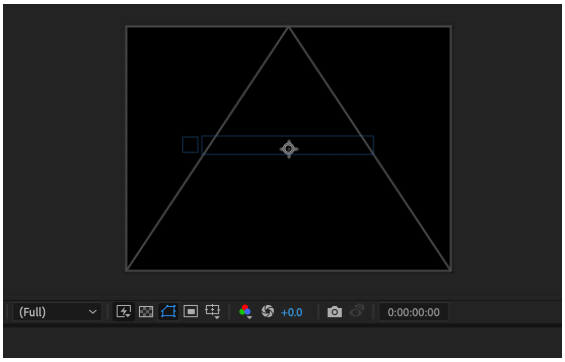
Set the duration of your *Composition Settings* to five seconds. Hit *Ok* to create your composition. Your composition will appear in your *Project* tab will a small film icon appearing to the left of it.

If you need to make changes to a composition after you have created it you can edit the setting by selecting the composition in the *Project* panel and by going to *Composition > Composition Settings*.

Timeline

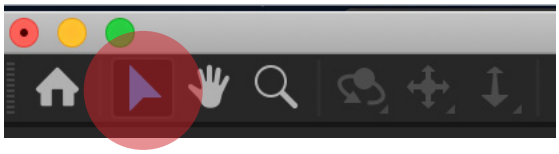
Now that we have a composition we can begin to animate our scene! Yay!

Begin by selecting *background-space.png* and drag it into the *Composition* window. Before letting go you'll see the wireframe of the image to show you where you are placing it. After dropping into the composition window you'll see that the image appears in the timeline at the bottom of the application as well.



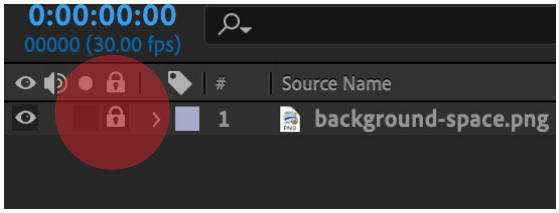
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To reposition the *Background-space.png* file use the *Selection Tool* found at the top left hand side of the program. The selection tool will allow you to place and resize the image inside the *Composition* panel.



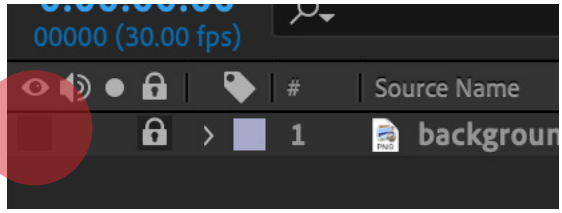
Once you have the *Background-space.png* image where you want it go to the *Timeline*. The timeline shows use when a piece of content starts and ends in our composition. You'll see the name *Background-Space.png* on the left hand side and a light blue bar that runs the length of the timeline, that means the image will appear for the entire composition.

To play it back our composition so far press the *spacebar* to play and press the spacebar again to pause the animation. You should see a red line moving from the left side of the timeline to the right, that is the *time marker*. At the moment this animation is still because it is an image, not a video.



Over on the far left hand side there are options for our image. Let's use the *lock* feature, which allows us to lock the content into place so we can not move or scale it by accident. To use the lock feature click the lock checkbox on the left side.

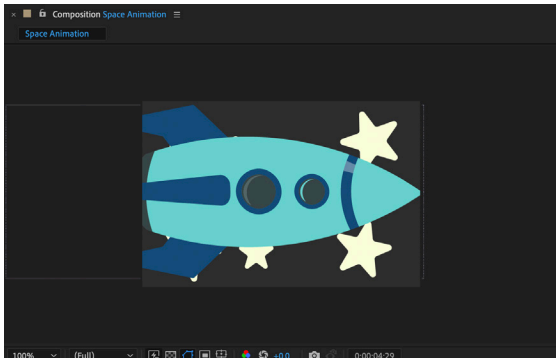
We can also turn the *Background-Space.png* file on and off in the timeline by using the *Visibility* toggle on the far left. The visiblity toggle allows you to turn on and off a layer in your timeline without deleting the layer itself.



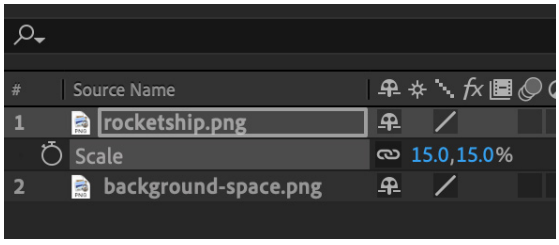
Animation

Now that we have a basic understanding of creating a composition in Adobe After Effects and adding files to it, let's learn how to animate with the program.

Begin by dragging the *rocketship.png* into the composition panel, the rocketship should appear in the timeline as well. If you look at the video example of the lab you'll see that the rocketship goes from the left side of the composition canvas to the right side of the canvas.

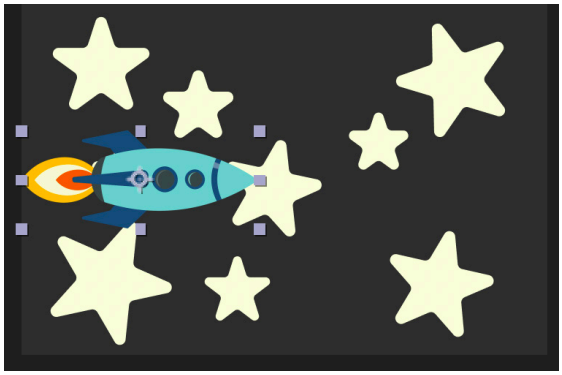


Before we animate the rocketship, you'll notice the rocketship is a bit large. You can resize the *rocketship.png* file by using the *Selection Tool* and using the built-in free transform to make the rocketship smaller. Another way to resize the rocketship is to select the *rocketship.png* layer in the timeline and press the letter *"S"* on the keyboard. When you do this you'll see that the word *"scale"* appears below the layer in the timeline and has two number values. The *scale* represents the size of your rocketship. The two values represent the width and height percentage of the rocketship image. The higher the numbers, the larger the image, the smaller the numbers, the smaller the image. Since



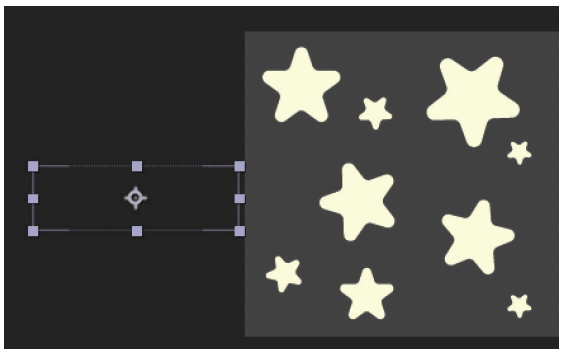
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we want the rocketship to be smaller let's scale the rocketship down. Change the value to 30% or 30% of the original image's dimensions, the result will be the rocketship image shrinks.

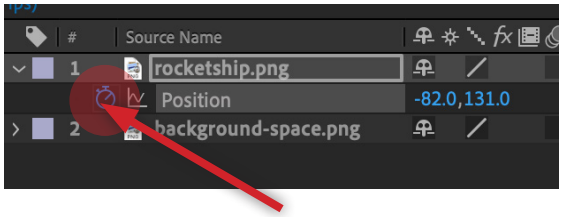


With the rocketship resized use the *Selection Tool* to place the image on the left side of th canvas in the *Composition* window. The rocket should be just to the left of the canvas so the viewer can not see it.

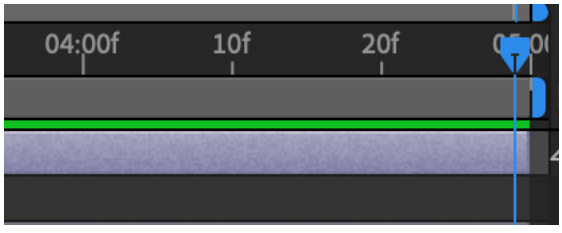
Now that our rocketship is in place we want to animate it going from the left side of the screen to the right side of the screen. To do this we will manipulate the *position* property of the layer. The *position property* represents where the image is within our canvas. With the rocketship.png layer selected in the timeline hit the *"P"* key on the keyboard to show the position property. You should see two values, these values are the horizontal or *x-position* and the vertical or *y-position* of the image. You can change the horizontal and vertical values here to change the position of the rocketship if you wish. Set the starting position of the rocketship. Make sure that the time marker is set to the beginning or 00:00:00:00 of the timeline.

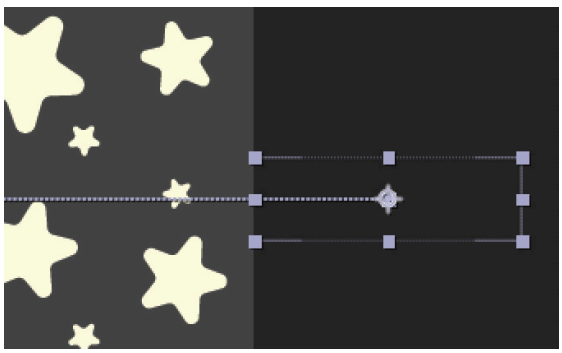


To the left of the position property you'll see a icon that looks like a stopwatch, this is the *keyframe* button. When a property like *position* has been keyframed it means any change to that property along the timeline will be recorded. To begin keyframing click the stopwatch icon, you will see the stopwatch turn blue and a small diamond shape appear in the timeline. This diamond represents a keyframe or a saved point in time for that property. In this case the saved property is a *position* of *-160, 250* at *00:00:00:00*.



Now that keyframing has been turned on, we do not need to turn it on again to create more keyframes, instead we will affect the layer to automatically create new keyframes. First make sure the *rocketship.png* layer is selected. Then take the time marker and move it to the end of the timeline to *00:00:05:00*. Use the *Selection Tool* to move the rocketship to the right side of the *Composition* window so that the rocketship is just to the right of the *Composition* window. If you view the timeline you will see an additional position keyframe has now been set at *00:00:05:00*. Take the timemarker and place it back at the beginning of the timeline, then hit the spacebar to place back your animation. The result is you should see the rocketship move from the left side of the *Composition* panel to the right hand side of the panel.





So what exactly happened here and how did the keyframes help? The keyframes represent saved positions in time. The keyframe at frame 1 saved that the rocketship started at *-160*

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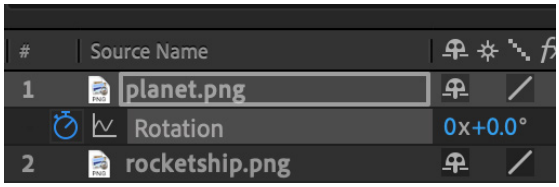
for the x-position. The second keyframe at the end of the animation saved the rocketship position at an x-position of *890*. In otherwords, After Effects was told that the rocketship x-position had to change from *-160* to *890* in five seconds. The result is the spaceship moved from the left side of the compositon screen to the right over five seconds.

Rotation

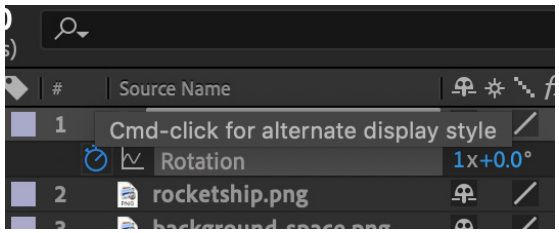
Now that we can animate a position of a layer, let's try animating the rotation of a layer. Drag the *Pwlanet.png* file from the *Project* panel into the *Composition* panel. Use the *Selection Tool* to move and scale the planet to the lower right hand corner of the *Composition* similar to the image on the right. Don't worry if it not exact.



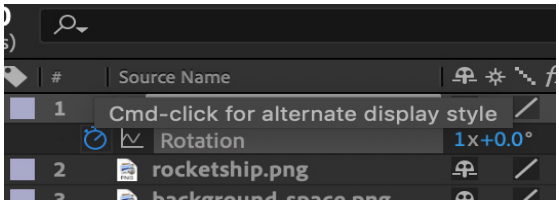
This time, instead of animating the position property, we will animate the rotation of the planet to make it spin. Move the time marker back to the beginning of the timeline. With the *planet.png* layer selected in the timeline hit the *"R"* key to show the *rotation* property. You'll see something that looks like this *0x + 0˚*. These numbers represent the number of times the layer rotates. Object rotates in measurements known as degrees and there are 360 degrees in a full rotation. The first number denotes the number of full rotations, the second number denotes degrees.



Set the *rotation* property to *0x +0˚*, if it not already. Like with the position property you'll see a stopwatch icon to allow us to keyframe this property. Click the stopwatch to begin keyframing the rotation property for the *planet.png* layer. Move the time marker to the end of the timeline. At the end of the timeline set the *rotation* property to *1x* or one full rotation. Take the timemarker and place it back at the beginning of the timeline, then hit the *spacebar* to place back your animation. The result is the planet should rotate one full rotation.



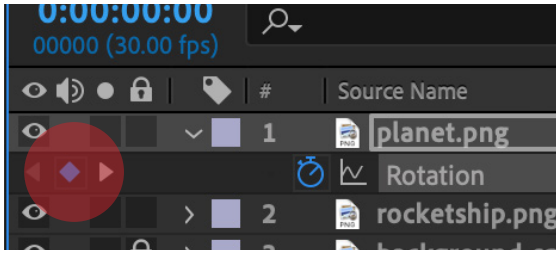
So again, what exactly happened here and how did the keyframes help? The keyframes represent saved rotation points in time. The keyframe at frame 1 saved that the planet rotation at 0˚. The second keyframe at the end of the animation saved the planet's rotation at *360˚* or *1x*. In otherwords, After Effects was told that the planet's rotation property had to change from *0* to *360* over five seconds. The result planet rotated one full rotation.



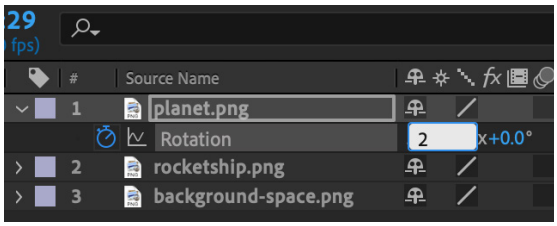
What if instead of one rotation we wanted to have the planet do two rotations? The answer is we would need to go back and edit the keyframes, remember the keyframes are the saved information After Effects uses to animate with. There are two ways to go back to an existing keyframe: **one** is take the time marker and place it upon the *diamond icon* in the timline. It can be tricky

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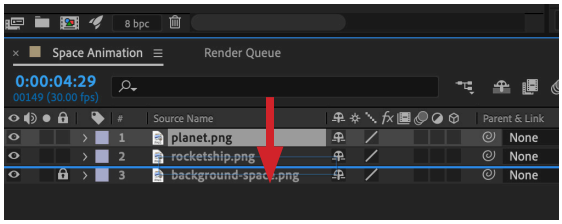
to be sure if you are actually over the diamond instead of near it. The easiest way to check is to view the keyframe indicator on the left side of the timeline panel. If the diamond icon turns blue then you have selected the keyframe. The other method of going between keyframes is to use the *arrow icons* in the timeline. These arrows on the far left let you jump from keyframe to keyframe within the layer.



Go back to the *rotation* keyframe at the end of the timeline. You should see the value of *1x* appear. Change this value to *2x*, this now has modified the value, which will modify the animation. Play the animation again from the start and you should see the planet do two full rotations.



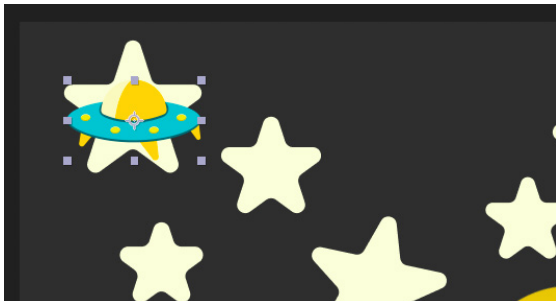
You may notice that the planet blocks your rocketship animation. This is because of the layer order in our timeline. Layers in Adobe After Effects are similar to those Adobe Photoshop where layers on top block layers below them. If the planet is blocking the rocketship that is because it is above it in timeline. To change the order take the planet layer and drag it below the rocketship layer, but above the *background-space.png* layer.



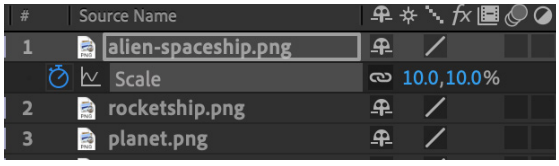
Save your project file.

Scale Property

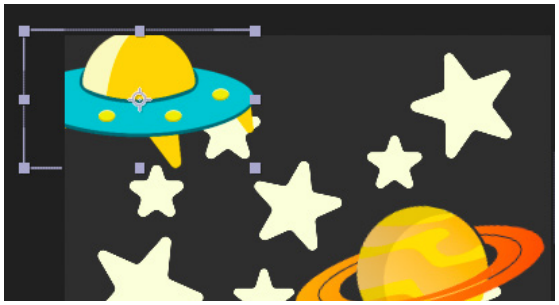
The last property we will play with today is the scale property. The scale property is the size of layer. Drag the *Alien-Spaceship.png* file from the *Project* panel into the *Composition* panel. Use the *Selection Tool* to place it at the top left hand side of the composition. Use the free-transform to resize it roughly to the size shown on the image on the right, don't worry it is not exact.



Set the time marker to the beginning of the timeline. Select the *Alien-Spaceship.png* layer and press *'S'* to view the scale property. Like with the position and rotation you should see a keyframe stopwatch to left of it. Click the stopwatch to begin keyframing the scale property of the *Alien-Spaceship.png* file.



Move the time marker to the end of the timeline. With the *Alien-Spaceship* layer selected in the timeline scale the alien spaceship photo larger using the *Selection Tool* or by using the scale property. Play back the timeline animation from the beginning and your *Alien-Spaceship.png* layer should scale in size overtime.



Save your project file.

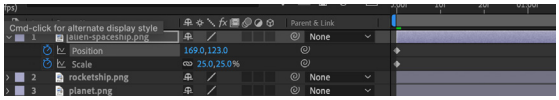
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Animate Multiple Properties



If you view the original animation example you'll see that the alien spaceship does not just scale, but it moves from the upper left corner of the composition to the middle while scaling. In other words the alien spaceship has two properties animating, the scale property and the position property. Layers in Adobe After Effects can have mulitple properties animating at the same time.

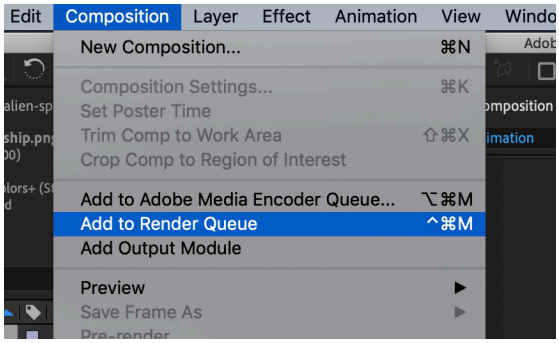
Move the time marker back to the beginning of the timeline. With the *Alien-Spaceship.png* layer selected press *"P"* to get the position property. Set the position property keyframe using the stopwatch icon. Use the time marker to move to the end of the animation. Use either the *Selection Tool* or the *Position* property values in the timeline to move the *Alien-Spaceship* to the middle of the composition. Play back the animation to make sure that the alien spaceship both scales and moves while animating.



When we animate multiple properties for a layer it can be nice to see ALL of the properties that we've keyframed. Select the Alien-Spaceship.png layer in timeline and press the *"U"* key on the keyboard. The *"U"* key will show every property that is keyframed for that layer. You should see two lines, each with their own keyframes, one for scale and one for position.

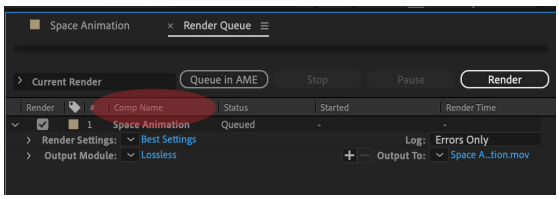
Save your project file.

Export Project



Whew! That was a lot. Now that we have are awesome animation let's export it so that we can share it with others. The After Effects Project file or *.aep* file only works within Adobe After Effects, since most people don't have After Effects installed on their computer we need to export our project to video format that we can share with others. After the video file has been exported it can be uploaded to services like YouTube or brought into video editing programs like Premiere Pro.

To export our animation make sure you are currently in our *Space Animation* compositon. Then go to *Compostion > Add to Render Queue*. The *Render Queue* allows us to set our settings for the outputted video file.



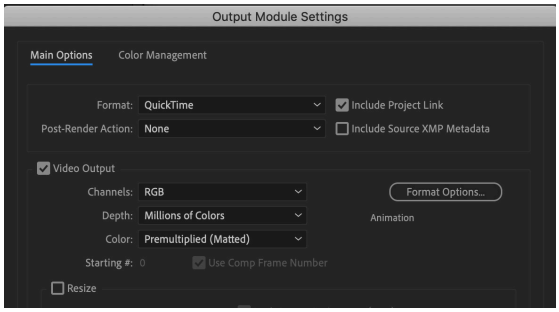
In the *Render Queue* look for *Comp Name* at the top. This tells us which composition is being exported, make sure it is set to *Space Animation*.

Under *Render Settings* make sure it is set to *"Best Settings".* This option is for the render quality, and we want the best settings of best quality.

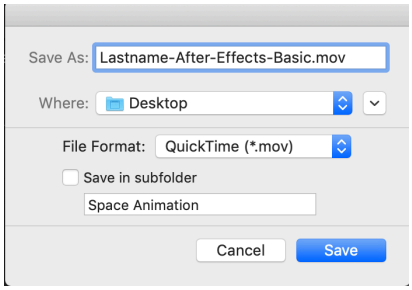
Under *Output Module* make sure it is set to *"Lossless."* Click on

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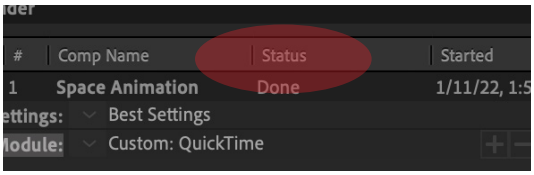
the word *Lossless* to open a new dialogue box. In the *Output Module Settings* under *Format*, make sure it is set to *Quicktime* or *AVI* video format. The *Output Module* allows us to set what type of video format we want to export as, for now we want Quicktime or AVI depending on the operating system of your computer. Hit *Ok* to return to the *Render Queue*.



Finall, use the *Output To* option to tell After Effects the name of exported video file and where it will be saved to. Label your file *lastname-After-Effects-Basics* and save it to the *Desktop* of your computer.



After the *Render Queue* settings have been set, click the *Render* button on the upper right hand side of the *Render Queue* bar. This will begin the process of outputting your video. The time is takes to output your video depends on the complexity of the project and how powerful your computer is. When the render is complete you will hear a sound and the *Render Queue* will show a *Status* of *Done*.



Submission

Upload your exported animation video format to your YouTube account. Publish the video and you will copy and paste the video URL to the submission page for the lab.

In addition to the final video url please upload the *lastname-after-effects-basics.aep* file to Canvas for grading. Please do NOT submit all of the files or videos as those are large files, instead just upload the *.aep* project file for grading purposes.

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